

每周科技深度周报

05-22 | 过去 7 天 | 主题式横向对比

本期导读

本周报聚合过去 7 天高分候选，按主题归并，保留英文产品名、项目名和论文名，并用中文说明作用、功能与对比风险。

已检查来源

GitHub Release、Google Developers Blog、arXiv cs.AI 论文、arXiv cs.CL 论文、arXiv cs.CV 论文、arXiv cs.LG 论文、arXiv stat.ML 论文

1. 重要论文与研究进展

来源: arXiv cs.AI 论文、arXiv cs.CV 论文、arXiv cs.LG 论文 | 类型: 主题汇总 | 主题: 重要论文与研究进展 | 时间: 过去 7 天, 4 条候选 | 分数: 9

是什么	本周该主题下的真实支撑条目：Cambrian-P: Pose-Grounded Video Understanding (arXiv cs.CV 论文, 05-21)：Camera pose matters. The position and orientation of each viewpoint define a shared spatial coordinate frame that relates observations across video frames. Yet this signal is larg...；Integrable Elasticity via Neural Demand Potentials (arXiv cs.LG 论文, 05-21)：We propose the Integrable Context-Dependent Demand Network (ICDN), a demand-first neural model for multiproduct retail demand. The model learns log-demand as a smooth, context-con...；Reducing Political Manipulation with Consistency Training (arXiv cs.AI 论文, 05-21)：Large language models (LLMs) exhibit systematic political bias across a variety of sensitive contexts. We find that LLMs handle counterpart topics from opposing political sides as...
主要作用	判断“重要论文与研究进展”方向是否出现值得调整技术选型、学习重点或产品路线的变化。
核心功能	支撑条目来自原始标题、摘要或 release notes：Camera pose matters. The position and orientation of each viewpoint define a shared spatial coordinate frame that relates observa...；We propose the Integrable Context-Dependent Demand Network (ICDN), a demand-first neural model for multiproduct retail demand. Th...
对比判断	原文出现 benchmark 相关线索：With a carefully designed sampling scheme, the model achieves substantial gains of 4.5-6.5% on spatial reasoning benchmarks such as VSI-Bench, generalizes across eight additional spatial and general video QA benchmarks, and, as a byproduct, achieves state of t；原文出现 benchmark, price 相关线索：The model learns log-demand as a smooth, context-conditioned function of log-prices, allowing elasticities to be derived exactly from the learned demand surface.
适合谁	适合跟踪相关论文方向、准备复现实验或寻找新研究 idea 的读者。

直达链接 1: [Cambrian-P: Pose-Grounded Video Understanding: https://arxiv.org/abs/2605.22819v1](https://arxiv.org/abs/2605.22819v1)

直达链接 2: [Integrable Elasticity via Neural Demand Potentials: https://arxiv.org/abs/2605.22820v1](https://arxiv.org/abs/2605.22820v1)

直达链接 3: [Reducing Political Manipulation with Consistency Training: https://arxiv.org/abs/2605.22771v1](https://arxiv.org/abs/2605.22771v1)

2. 推理性能与基础设施

来源: GitHub Release、Google Developers Blog、arXiv cs.AI 论文、arXiv cs.CV 论文 | 类型: 主题汇总 | 主题: 推理性能与基础设施 | 时间: 过去 7 天, 9 条候选 | 分数: 9

是什么	本周该主题下的真实支撑条目：DeltaBox: Scaling Stateful AI Agents with Millisecond-Level Sandbox Checkpoint/Rollback (arXiv cs.AI 论文, 05-21)：LLM-powered AI agents require high-frequency state exploration (e.g., test-time tree search and reinforcement learning), relying on rapid checkpoint and rollback (C/R) of the comp...；WorldKV: Efficient World Memory with World Retrieval and Compression (arXiv cs.CV 论文, 05-21)：Autoregressive video diffusion models have enabled real-time, action-conditioned world generation. However, sustaining a persistent world, where revisiting a previously seen viewp...；A2UI v0.9: The New Standard for Portable, Framework-Agnostic Generative UI (Google Developers Blog, 05-22)：A2UI v0.9 introduces a framework-agnostic standard designed to help AI agents generate real-time, tailored UI widgets using a company’ s existing design system. This update simplif...
主要作用	判断“推理性能与基础设施”方向是否出现值得调整技术选型、学习重点或产品路线的变化。
核心功能	支撑条目来自原始标题、摘要或 release notes：LLM-powered AI agents require high-frequency state exploration (e.g., test-time tree search and reinforcement learning), relying...；Autoregressive video diffusion models have enabled real-time, action-conditioned world generation. However, sustaining a persiste...
对比判断	原文出现 latency 相关线索：Existing mechanisms duplicate the entire state, causing hundreds of milliseconds to seconds of latency per C/R, which severely bottlenecks deep search and large-scale fan-outs.；原文出现 throughput, cost 相关线索：Full KV-cache attention preserves this consistency but breaks real-time constraints: memory footprint and attention cost grow linearly with rollout length.
适合谁	适合构建智能体、工具调用、开发者平台和自动化工作流的团队。

直达链接 1: [DeltaBox: Scaling Stateful AI Agents with Millisecond-Level Sandbox Checkpoint/Rollback: https://arxiv.org/abs/2605.22781v1](https://arxiv.org/abs/2605.22781v1)

直达链接 2: [WorldKV: Efficient World Memory with World Retrieval and Compression: https://arxiv.org/abs/2605.22718v1](https://arxiv.org/abs/2605.22718v1)

直达链接 3: [A2UI v0.9: The New Standard for Portable, Framework-Agnostic Generative UI: https://developers.googleblog.com/a2ui-v0-9-generative-ui/](https://developers.googleblog.com/a2ui-v0-9-generative-ui/)

3. 智能体与开发者平台

来源: GitHub Release、arXiv cs.AI 论文、arXiv cs.CL 论文 | 类型: 主题汇总 | 主题: 智能体与开发者平台 | 时间: 过去 7 天, 6 条候选 | 分数: 9

是什么	本周该主题下的真实支撑条目: Agentic CLEAR: Automating Multi-Level Evaluation of LLM Agents (arXiv cs.CL 论文, 05-21): Agentic systems are becoming more capable: agents define strategies, take actions, and interact with different environments. This autonomy poses serious challenges for overseeing...; mlflow/mlflow v3.13.0rc0 (GitHub Release, 05-22): We're excited to announce MLflow 3.13.0rc0, which deepens agent observability, tightens permissions, and broadens deployment options: **Major New Features**: - **RBAC Phase 2 + Ad...; Advancing Mathematics Research with AI-Driven Formal Proof Search (arXiv cs.AI 论文, 05-21): Large language models (LLMs) increasingly excel at mathematical reasoning, but their unreliability limits their utility in mathematics research. A mitigation is using LLMs to gene...
主要作用	判断“智能体与开发者平台”方向是否出现值得调整技术选型、学习重点或产品路线的变化。
核心功能	支撑条目来自原始标题、摘要或 release notes: Agentic systems are becoming more capable: agents define strategies, take actions, and interact with different environments. This...; We're excited to announce MLflow 3.13.0rc0, which deepens agent observability, tightens permissions, and broadens deployment opti...
对比判断	原文出现 benchmark 相关线索: In our experiments on four benchmarks, seven agentic settings, and tens of thousands of LLM calls, we show that Agentic CLEAR produces; 暂无可信公开对比数据; 标题和摘要中没有提供统一 benchmark、价格、延迟或吞吐指标。
适合谁	适合构建智能体、工具调用、开发者平台和自动化工作流的团队。

直达链接 1: [Agentic CLEAR: Automating Multi-Level Evaluation of LLM Agents: https://arxiv.org/abs/2605.22608v1](https://arxiv.org/abs/2605.22608v1)

直达链接 2: [mlflow/mlflow v3.13.0rc0: https://github.com/mlflow/mlflow/releases/tag/v3.13.0rc0](https://github.com/mlflow/mlflow/releases/tag/v3.13.0rc0)

直达链接 3: [Advancing Mathematics Research with AI-Driven Formal Proof Search: https://arxiv.org/abs/2605.22763v1](https://arxiv.org/abs/2605.22763v1)

4. 检索增强与向量模型

来源: arXiv stat.ML 论文 | 类型: 主题汇总 | 主题: 检索增强与向量模型 | 时间: 过去 7 天, 1 条候选 | 分数: 9

是什么	本周该主题下的真实支撑条目: Guiding Multi-Objective Genetic Programming with Description Length Improves Symbolic Regression Solutions (arXiv stat.ML 论文, 05-21): Symbolic regression with genetic programming (GPSR) may suffer from overfitting and structural bloat, especially when noise is present. In this paper we evaluate description lengt...
主要作用	判断“检索增强与向量模型”方向是否出现值得调整技术选型、学习重点或产品路线的变化。
核心功能	支撑条目来自原始标题、摘要或 release notes: Symbolic regression with genetic programming (GPSR) may suffer from overfitting and structural bloat, especially when noise is pr...
对比判断	原文出现 benchmark 相关线索: We implement DL using a Fisher-information-based parameter encoding and compare it to AIC and BIC across multiple datasets, including noisy synthetic benchmarks and real-world regression problems.
适合谁	适合做企业知识库、搜索、RAG、向量数据库和多语言检索的团队。

直达链接: [Guiding Multi-Objective Genetic Programming with Description Length Improves Symbolic Regression Solutions: https://arxiv.org/abs/2605.22374v1](https://arxiv.org/abs/2605.22374v1)

备注: 周报优先复用同一套低成本抓取和评分逻辑。